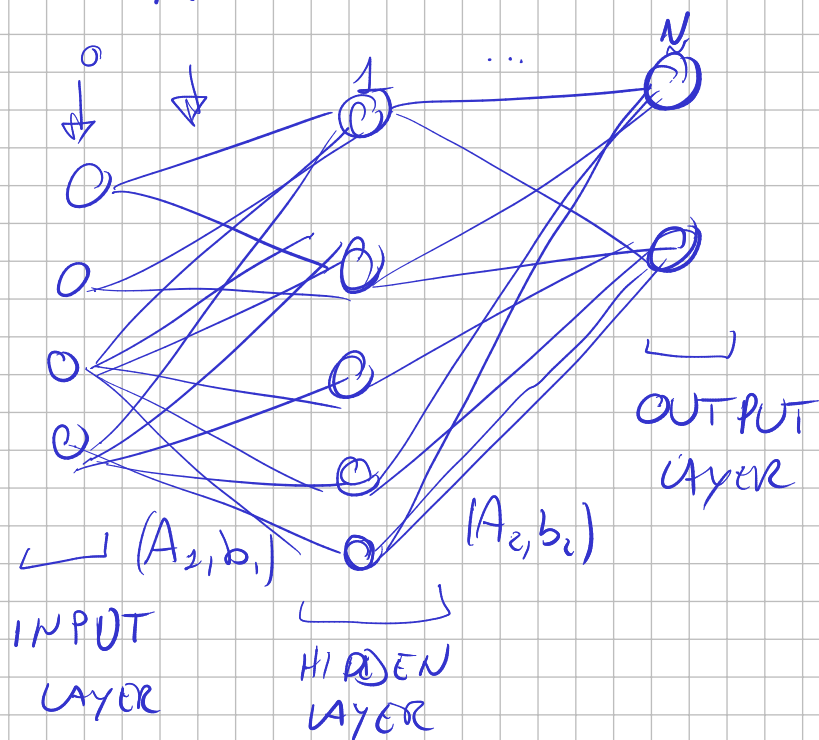


DEEP NEURAL

MULTI LAYER PERCEPTION



HIDDEN layer $\rightarrow \bigcirc = \text{FUNCTIONI NON LINEARI}$
ACTIVATION FUNCTIONS

$$NN(x) = \sigma(A_N + \dots \sigma(A_2 \sigma(A_1 x + b_1) + b_2) + \dots) + b_N$$

$$\begin{cases} A_i \in \mathbb{R}^{m_i \times m_{i-1}} \\ b_i \in \mathbb{R}^{m_i} \end{cases}$$

σ ACTIVATION FUNCTIONS

N = NUMBER OF LAYERS

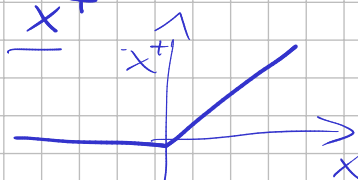
m_i = # DI Nodi del layer i -esimo

$$\sigma: \mathbb{R} \rightarrow \mathbb{R}$$

$$\sigma: \mathbb{R}^m \rightarrow \mathbb{R}^m \quad \forall m \text{ COMPONENT-WISE}$$

ReLU
RECTIFYING LINEAR UNIT

$$\sigma(x) = x^+$$



ESEMPLI DI ACTIVATION FUNCTIONS

• ReLU (Rectifying linear unit) $R(x) = x^+ = x \mathcal{H}_{(0, +\infty)}$

NOTO SEMPLICE!

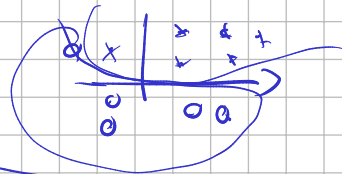
- Leaky Relu $(x) = ax \chi_{(-\infty, 0)}$
 $+ x \chi_{(0, \infty)}$
 $0 < a < 1$



- tanh(x) ✓
- softplus $(x) = \log(1 + e^x)$



NN → CLASSIFICAZIONE
 → REGRESSIONE



1

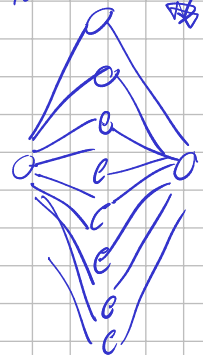
FUNZIONE TARGET f che vogliamo approssimare

$$\min_{\theta} \|NN_{\theta}(\cdot) - f(\cdot)\|$$

$$NN(x, \underbrace{A_1, b_1, A_2, b_2, \dots, A_N, b_N}_{=: \theta \in \mathbb{R}^M})$$

- DATI UNA f CONTINUA SU UN COMPATTO $C \subseteq \mathbb{R}^{n_0}$
 UNA NN CON 1 HIDDEN LAYER E n_1 NODI NEZZA HIDDEN LAYER
 $\forall \epsilon > 0 \quad \exists n_1 \in \mathbb{N} : NN$

$$\max_{x \in C} \min_{A, b} |NN(x) - f(x)| < \epsilon$$



DEFINIAMO FUNZIONE COSTO

$$\{(x_i, y_i)\}_{i=1}^{N_{\text{train}}}$$

$$y_i = f(x_i)$$

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_{N_{\text{train}}} \end{pmatrix}$$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_{N_{\text{train}}} \end{pmatrix}$$

$$N_{\text{train}} \gg 1$$

$$f: \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_N}$$

$$\mathcal{L}(\theta) = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} (NN_{\theta}(x_i) - y_i)^2$$

MSE = MEAN SQUARED ERROR

$$\mathcal{L}: \mathbb{R}^M \rightarrow \mathbb{R}^+$$

θ
" $A_1, b_1, \dots, A_L, A_N, b_N$

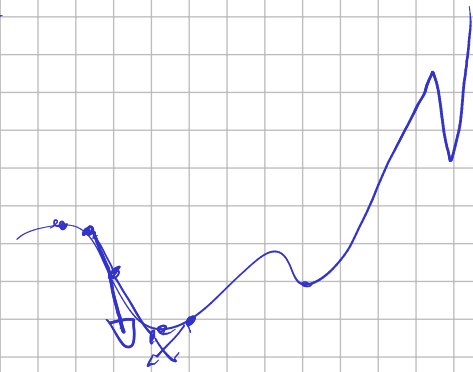
MINIMIZE THE LOSS FUNCTION $\mathcal{L}$

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$$

$$\theta_i^{n+1} = \theta_i^n - \eta \boxed{\nabla_{\theta_i} \mathcal{L}(\theta^n)}$$

DISCERN THE GRADIENT

η LEARNING RATE $\approx 10^{-3}$



GRADIENTE

VARIABLE THE LOSS FUNCTION!

FORWARD PASS

$$\begin{cases} h_0 := x \\ f_k := A_k h_{k-1} + b_k \\ h_k := \sigma(f_k) \end{cases} \quad k=1, \dots, N$$

$$NN(x) := h_N$$

$$\mathcal{L}(\theta) = \frac{1}{N_{\text{tr}}} \sum_{i=1}^{N_{\text{tr}}} (NN(x_i) - y_i)^2 = \frac{1}{N_{\text{tr}}} \|h_N - y\|_2^2$$

BACK PROPAGATION

$$\nabla_{\theta} \mathcal{L} = ?$$

$$\theta = (A_1, b_1, A_2, \dots, b_N)$$

$$g_{h_N} := \nabla_{h_N} \mathcal{L} = \frac{2}{N_{\text{tr}}} (h_N - y)$$

$$g_{f_N} := \nabla_{f_N} \mathcal{L} = \underbrace{\nabla_{f_N} h_N}_{= g_{h_N}} \cdot \underbrace{\nabla_{h_N} \mathcal{L}}_{= g_{h_N}} = \sigma'(f_N) \cdot g_{h_N}$$

$$\underline{g_{b_N}} = \nabla_{b_N} \mathcal{L} = \underbrace{\nabla_{b_N} f_N} \cdot \underbrace{\nabla_{f_N} \mathcal{L}} = \nabla_{f_N} \mathcal{L}$$

$$\underline{g_{A_N}} = h_{N-1} \cdot g_{f_N}$$

STOCHASTIC GRADIENT DESCENT

MOR + NN

$$u(\mu) \in \mathbb{R}^{N_h}$$

$$\text{POD} \rightarrow \psi \in \mathbb{R}^{N_h \times l}$$

$$\underline{u(\mu)} \approx \underbrace{\psi}_{\mathbb{R}^l} \cdot \underbrace{u^l(\mu)}_{\mathbb{R}^l}$$

μ PARAMETRO
(TEMPO / FISICA)

Se con POD-GALERKIN PROIEZIONE DI EQUAZIONI

PER AVERE $u^l(t) + \dots$

$$\left[\begin{array}{c|c} \mu & \text{POD} \\ \hline t & \end{array} \right] \begin{array}{c} u^l(\mu) \\ u^l(t) \end{array}$$

$$u^l: \underbrace{\mathbb{R}}_t \rightarrow \mathbb{R}^l$$

$$\boxed{\mu = t}$$

OFFLINE 1. POD (ψ)

$$\psi = \{u(t_i)\}_{i=1}^{N_{\text{TRAIN}}}$$

2. $\psi \in \mathbb{R}^{N_h \times l}$

3. $\{t_i, u^l(t_i)\}_{i=1}^{N_{\text{TRAIN}}}$

$$\underline{u^l(t_i)} = \psi^T u(t_i)$$

4. MINIMIZZARE $\sum_{i=1}^{N_{\text{TE}}} \|NN(t_i) - u^l(t_i)\|_2^2 \perp_{N_{\text{TRAIN}}}$
 $\Theta = A_1, b_1, A_N$

ONLINE

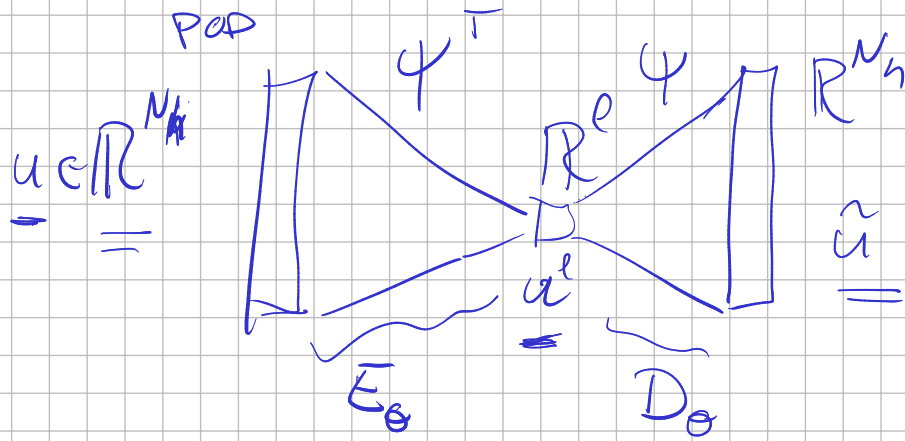
1. NUOVO PARAMETRO / TEMPO t $NN(t) \approx u^l(t)$

2. $u(t) \approx \psi \cdot NN(t)$

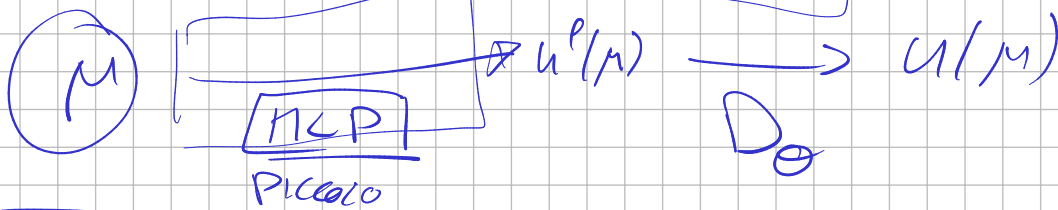
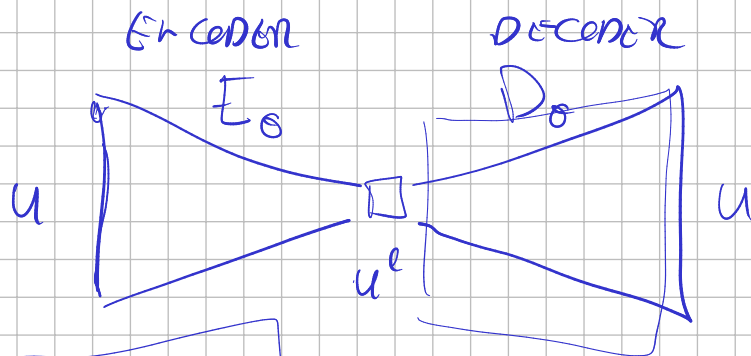
AUTOENCODER

VERSIONE NON LINEARE

PAD



$$\mathcal{L}(\theta) = \frac{1}{N_{tr}} \sum_{i=1}^{N_{tr}} \| u(t_i) - \underline{D_\theta(E_\theta(u(t_i)))} \|^2$$



CARLOR
FRESCA

$$NN(m) \approx u(m) \in \mathbb{R}^{N_h}$$

$$M \longrightarrow u \in \mathbb{R}^{N_h}$$

$$\{m_i, u(m_i)\}$$

$$\mathcal{L}(\theta) = \sum_{i=1}^N \| NN(m_i) - \underline{u(m_i)} \|^2$$

$$u(m_i) : \mathbb{R} \rightarrow \mathbb{R}^{N_h}$$

$$u(\mu, x) : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\mathcal{L}(\theta) = \frac{1}{N_x N_\mu} \sum_{i=1}^{N_x} \sum_{j=1}^{N_\mu} |NN(x_i, \mu_j) - u(x_i, \mu_j)|^2$$